MS EXCEL PRACTICAL

Student Result Management System

Step-by-step practical guide

Software	Microsoft Excel (2016, 2019, 2021, Microsoft 365; ribbon labels may vary)
Tabs covered	Home • Insert • File
Level	Beginner / Computer practical class
Suggested file name	Student_Result.xlsx

Aim: Create a formatted student marksheet, calculate totals and percentages, display results, insert tables and charts, and save/print the workbook.

What you will practise

• Entering data and using formulas.

• Formatting cells, headings, borders and conditional formatting.

• Inserting an Excel table, column chart, pie chart, picture, shape, text box and hyperlink.

• Saving, printing and exporting the workbook.

Prepared as a classroom practical worksheet

PART 1 — Create the worksheet

1. Open Excel and enter the data

Step 1. Open Microsoft Excel and choose **Blank workbook**.

Step 2. Click cell **A1** and type: **STUDENT RESULT MANAGEMENT SYSTEM**.

Step 3. Leave row 2 blank. In row 3, enter the column headings shown below.

Step 4. Enter the eight student records in rows 4 to 11. Leave the Total, Percentage and Result cells blank for now.

Enter this data exactly:

Roll No.	Student Name	English	Hindi	Computer	Total	Percentage	Result
101	Rahul	78	85	90	—	—	—
102	Priya	88	92	95	—	—	—
103	Amit	65	70	75	—	—	—
104	Neha	92	89	96	—	—	—
105	Rohit	35	42	38	—	—	—
106	Pooja	80	76	85	—	—	—
107	Mohan	28	32	25	—	—	—
108	Anjali	95	94	98	—	—	—

Important: These instructions use row 3 for headings and rows 4–11 for students. Formulas below are written for this exact layout.

2. Calculate Total, Percentage and Result

Step 5. Click cell **F4**, type `=SUM(C4:E4)`, then press Enter.

Step 6. Click cell **G4**, type `=F4/300*100`, then press Enter.

Step 7. Click cell **H4**, type `=IF(MIN(C4:E4)>=33,"Pass","Fail")`, then press Enter.

Step 8. Select F4:H4. Drag the small square at the lower-right corner of the selection down to row 11 to copy all three formulas.

Result rule: This worksheet treats a student as Pass only when each subject mark is at least 33. Adjust the formula if your school uses a different passing rule.

PART 2 — Use the Home tab

3. Format the title and data

Step 9. Select cells **A1:H1**. On **Home**, click **Merge & Center**.

Step 10. With A1 selected, make the title **Bold**, set font size to **16**, choose a font colour, and apply a fill colour.

Step 11. Select A3:H3. Click **Bold**, choose a contrasting fill colour, and set horizontal alignment to Center.

Step 12. Select A3:H11. Choose **Borders** → **All Borders**.

Step 13. AutoFit columns: select columns A:H, then choose **Home** → **Format** → **AutoFit Column Width**. You may also double-click a column boundary.

Step 14. Select G4:G11 and apply Number format with 2 decimal places, or Percentage format if you prefer the values to display with a percent sign. If using Percentage format, change the formula in G4 to `=F4/300` before filling down.

4. Highlight low marks and format as a table

Step 15. Select C4:E11. Choose **Home** → **Conditional Formatting** → **Highlight Cells Rules** → **Less Than**. Enter **40** and choose a red highlight style.

Step 16. Select A3:H11 and choose **Home** → **Format as Table**. Pick a style and confirm **My table has headers**.

Step 17. Try **Sort & Filter**: use the Student Name heading's filter arrow to sort A to Z. To return to roll-number order, sort Roll No. smallest to largest.

Step 18. Practise **Find** (Ctrl+F), **Copy** (Ctrl+C), **Paste** (Ctrl+V), and **Undo** (Ctrl+Z).

Home tab skills covered: Clipboard, Font, Alignment, Number, Styles, Cells and Editing groups.

PART 3 — Use the Insert tab

5. Insert an Excel table

Step 19. If you did not format the range as a table in Step 16, select A3:H11 and choose **Insert** → **Table**.

Step 20. Tick **My table has headers**, click OK, and choose a suitable table style.

6. Insert a column chart

Step 21. Select the non-adjacent Student Name and subject-mark columns. Easiest method: select B3:E11, which includes Student Name and all three subjects.

Step 22. Choose **Insert** → **Insert Column or Bar Chart** → **Clustered Column**.

Step 23. Click the chart title and type **Subject-wise Student Marks**. Move the chart below or beside the table.

7. Insert a pie chart

Step 24. Select B3:B11. Hold Ctrl (Windows) or Command (Mac), then select F3:F11. Alternatively, create a small two-column helper table with Student Name and Total.

Step 25. Choose **Insert** → **Pie Chart**, then select a 2-D Pie chart.

Step 26. Change the chart title to **Total Marks of Students**. Use Chart Elements to add data labels if available.

PART 4 — Insert supporting objects

8. Add a picture, shape and text box

Step 27. Click a blank area of the worksheet and choose **Insert** → **Pictures**. Select a suitable school logo or classroom image from your computer. Resize it using the corner handles.

Step 28. Choose **Insert** → **Shapes**, select a rectangle, and draw it in a blank area. Type **Prepared by: [Your Name]** inside it.

Step 29. Choose **Insert** → **Text Box**, draw the box, and type **Academic Report**. Format it as needed.

9. Add a hyperlink

Step 30. Select an empty cell, such as A13, and type **School Website**. With the cell selected, choose **Insert** → **Link** (or press Ctrl+K / Command+K) and enter your school's official website address.

10. Add a header, footer and page number

Step 31. Use **Insert** → **Text** → **Header & Footer** if your Excel version shows this option. In the header, type the school name.

Step 32. Click the footer area and insert the page number using the Header & Footer tools. Return to Normal view when finished.

11. Optional: add a Sparkline

Step 33. If your version supports Sparklines, select an empty cell for the first student, choose **Insert** → **Sparklines** → **Line**, set Data Range to C4:E4 and Location Range to the chosen cell, then click OK. Repeat for other students if desired.

Note: The Insert ribbon differs slightly between Windows, Mac and Excel versions. If an option is not visible, use the ribbon search box or skip that optional item.

PART 5 — Use the File tab

12. Save your workbook

Step 34. Choose **File** → **Save As**.

Step 35. Select a folder, enter **Student_Result.xlsx** as the file name, and click Save.

Step 36. Press Ctrl+S (Windows) or Command+S (Mac) regularly to save changes.

13. Review workbook information and reopen it

Step 37. Choose **File** → **Info** (if available) to review workbook properties and protection options.

Step 38. Choose **File** → **Open** to practise opening a previously saved workbook. Return to your practical file.

PART 6 — Print, export and submit

14. Set up the print layout

Step 39. Choose **File** → **Print** to open Print Preview.

Step 40. If the table is too wide, choose **Page Orientation** → **Landscape**.

Step 41. Set suitable **Margins**. Under scaling, choose **Fit All Columns on One Page** if required.

Step 42. Check that headings, marks and totals are readable. Select the intended printer and print only if instructed.

15. Export a PDF copy

Step 43. Choose **File** → **Export** or **File** → **Save As**, depending on your version.

Step 44. Select **PDF** as the file type, choose a location, and save as **Student_Result.pdf**.

Step 45. Open the PDF and confirm the table and charts are not cut off. If needed, adjust scaling and export again.

16. Final submission checklist

■ All student data entered correctly

■ Total, Percentage and Result formulas copied to all students

■ Title, headings, borders and column widths formatted

■ Low marks highlighted using Conditional Formatting

■ Table and charts inserted

■ Picture, shape, text box and hyperlink added

■ Header/footer or page number attempted where available

■ Workbook saved as Student_Result.xlsx

■ Print Preview checked and PDF exported if required

Viva questions

1. What is the difference between a workbook and a worksheet?
2. What does Merge & Center do?
3. Which function calculates total marks?
4. What is Conditional Formatting?
5. How do you insert a chart?
6. What is the difference between Save and Save As?
7. What is Print Preview used for?
8. What is a hyperlink?

Practical submission: Submit Student_Result.xlsx and, if requested by your instructor, Student_Result.pdf.